Strong senior DS with…
Data science + data engineering fullstack background expanding to gen ai, building RAG applications,
Open to work in Toronto, but interested in NYC/NJ opportunities.

INTRODUCTION:

1. **SCHOOL, CURRENTLY DS BMO, INTERSECTING X** **ML mod/bizip**

2. **2 LENSES:** HANDS ON & DS CONS, turn biz prob to DS SOLNS-> **results**
Action, measurable

3. **FGF BRANDS:** FOOD PROD. OBJ DET, CV, TF, defic in manu proces

4. **TRANSITION, WHY BMO** CLIENT B, IMPACT ON STRAT

5. **BMO SPLIT:** ML INFRA 1st hire, featstore, autML NOW XSELL, NLP,UOFT

6. **Look to take NEXT STEP @ DD:** my TECH EXE + STAKE FACE CNSLT

7. **where** FOCUS ON **IMPACT, PRIO. URGENCY, OUTCOMES, RESULT**

8. **WORK W SMART PPL,** SHARE IDEAS, COLLAB W, LEARN FROM, FUEL CURIOSITY

<u>MULTI-CLASS:</u>

SITUATION: sev LOB,

- IL self invmt, lose mrktshr/clients to WS,QT
- biz **dk**. MY IDEA, I **said** if I can PRED WHERE
- DESIGN TAILORED OFFERS to where CLIENTS TRANSFER ASSETS to

TASK:

- DISTINGUISH WHERE PPL move
- Tailor strategy w/ PERSONALIZED OFFERS to help retain these clients.
- BUILD MULTICLASS for clients likely to attr
- USED DEMOG, BEHVRAL, TXN HIST (in/out)

ACTION:

- ENGINEERED KEY FEATURE (3/6/9mo), TXNS
- ENGINEERED TGT (NLP STR MTCH)
- RF MODEL

RESULT:

- **PRESENTED** DECK STRATEGY TEAM
- MONTHLY **LEADS** TO STRAT FOR MRKTING Q CAMPAIGNS
- BEYOND PRED, **PROFILING**, BUILT **PERSONAS**
- EX: WS (age, lowAUM, airpod)
- EX: TD (age, AUM, txn#, fees incurred, discnt)

ALL IN ALL: **MODEL OUTPUTS + PROFILING** clients clients

present ledshp **NOT ONLY WHERE**, **GO**, but **WHO**, **ARE**

QUESTIONS:

Q: **explain to stakeholders technical things in a non-technical way**, and be able to communicate effectively despite this hurdle?

Q: Challenge you faced in explaining complex model results to non-tech stakeholders? Maybe had to adjust style to get buy-in, tell me how you did it.

- Comes to model metrics and performance, **work backwards.**
- Communicate and align with stakeholders on **priorities** and **cost** (understand the **penalty** of incorrectly offering a TD offer, vs WS offer)
- Instead of diving into prec/recall, understanding what is **best for stakeholders**.
- They wanted to see **WHO these clients are and HOW to act on them.**
- Turn precision scores into **biz terms** "predict 8 of every 10 correctly". Understand their needs
- Helping me create threshold minimums to avoid forceful pred, focusing on precision>rec.
- Communicating model results in **English** like **PERSONAS** and **profiles** over metrics.
- **Iterative feedback loops** to create ongoing path of communication, so check-ins and things can be **tweaked** along the way.
- **Continuous dialogue**.

Q: ROADBLOCK?

- Orig. looked into data, 20+ institution strmtch.
- To **handle data SPARCITY**, K-MEANS CLUSTER
- 4-5 features defining client behavior
- Showed brokerages like MANULIFE/ASSANTE
- CIBC, RBC, HSBC together. WS, TD on own.
- GIVES **EVIDENCE** for **COLLAPSING** SMALLER FREQ INSTITUTIONS **TG**.
- **TRICKY**
- Biz "want RBC". "difficult, customers similar, act same, hard for MODEL DISTINGUISH".
- Did it anyway!
- Didn't want to FLOOD FALSE POS, **MITIGATE** this: if **not superconf**, label to **OTHER FI**.
- **INCR** PRECISION RBC, **ULTRASELECTIVE**.
- **What they wanted vs. what data showed.**

Q: **THRESHOLD** DECISIONS?

- Class chosen is highest prbility, but added **UNKNOWN** to avoid **WEAK SIGNALS**.
- **AVOID** FORCEFUL PRED, @ COST OF BIZ!
- TD offer $$$ > WS offer, **tighter leash on TD.**
- Calculated PRECISION 0.5,0.6,0.7, to see **MISCLASS RATE + COVERAGE %.**
- **Distribution of prbility to see natural gap in misclass rate.**

- Look for **DROP OFF**, SWEET SPOT, **PER CLASS BASIS**, while keeping coverage% respectable.

Q: **MEASURE SUCCESS** OF MODEL?

- PRECISION > RECALL since "UNKNOWN".
- NOT accuracy, data imbalance, only prio.TD.
- BACKTESTING RESULTS TO BIZ.
- CONSTANT **FEEDBACK LOOP** w biz as PRIORITIES CHANGE THRU TIME (new offers)

BACKTESTING:

- **Take pred,** look up to **3m later OUTFLOWS**, to see **how many** PRED TO X, ACC WENT TO X.
- **Compare PRECISION W TESTING**. Share w biz.
- Look @ CONFUSION MATRIX, for any MAJOR MISCLASSIFICATION TRENDS.
- CHECK FOR **INCONSISTENT MISSCLASS ERRORS** in specific categories(X always wrong)

Q: CONFUSION MATRIX?

- Table shows n x n for n classes.
- Col = PRED, row = ACTUAL.

Q: RANDOM FOREST

- ENSEMBLE LEARNING METHOD
- Builds up many decision trees on RANDOM SUBSETS of features/data, random features @ each node. **Each tree makes splits on features**
- **Averaging their results** for classification.

- Forest pred = majority vote or avg’d PR()s
- Built-in feature importance by looking HOW MUCH each feature contributes to reducing impurity across ALL TREES (improving preds) across each tree split.
- LARGEST REDUCTION = most important.
- Good **MIDDLEGROUND** btwn PERFORMANCE + SIMPLICITY, robustly **handles** outlier/nulls.
- Used SMOTE

Q: CLUSTERING

- Unsupervised learning method grouping data points on SIMILARITY
- No pre-defined labels
- K CLUSTERS picked, assigns each data point to NEAREST CENTROID.
- Iteratively UPDATES CENTROIDS till assignments STABILIZE.
- MINIMIZE distance within clusters, MAXIMIZE separation btwn them!

⭐ SITUATION

- Had **200K+ unstructured call logs** from client call centers (frontline) with no clear use.
- Frontline teams were **manually reviewing calls** to detect insights.
- Your manager asked if you had **ideas on how to put this data to use**.
- You proposed summarizing the calls to:
  - Help frontline better understand **client intent**
  - Surface **cross-sell opportunities**
  - Enable clear **call-to-action insights** on the dashboard

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⭐ TASK

- Build an **NLP pipeline** to surface actionable insights from call logs.
- Develop an **automated summarization tool** to extract the **most recent 3 calls per client**, in bullet-point form, and store them for use in the frontline dashboard.

⭐ ACTION

- **Data ingestion:**
  - Pulled call logs from **Salesforce**
- **Preprocessing with NLP:**
  - Used **NLTK** for tokenization and POS tagging
  - Removed **named entities** (e.g., advisor names, tickers) via manual/regex cleaning
  - Applied **lemmatization** and custom cleaning (e.g., normalizing terms like “wd” → “withdraw”)
  - Filtered for **finance-related verbs and nouns** to keep relevant content
- **Summarization:**
  - Used **TextRank** (via Sumy) to extract key phrases
  - Focused on **action-oriented summaries** (calls that ended with specific outcomes)
  - Added **post-summarization filtering** and finalized the **most recent 3 logs per client**
- **Aggregation & storage:**
  - Summaries were **stored in a dictionary**, keyed by client ID, with a deque of size 3
- **Preliminary analysis (for insights):**
  - Used **n-grams** from NLTK, **word frequency charts**, **word clouds**, and **co-occurrence graphs**
  - Applied **TF-IDF** to extract category-specific key terms
  - Used **K-Means clustering** to group calls by topic/client type

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⭐ RESULT

- Summarization tool was **rolled out to 500+ IL frontline employees**
- **97%** of logs had **actionable summaries of up to the last 3 client calls**
- **Manual review time reduced by 80%** (from 3–5 minutes to under 1 minute)
- Stakeholders validated summaries and were able to:
  - **Search summaries directly from the dashboard**
  - **Port logs to Salesforce**
  - Use them for **cross-sell prompts and client follow-ups**

⚠️ ROADBLOCK

- Original LLM solution required lengthy **6+ month stakeholder approvals**
- Pivoted to a **lighter-weight, local TextRank approach** (Python-based) to deliver fast
- Engaged with **GMs** to get early feedback and improve UX before broader rollout

Fast paced, work directly with group head, AVP, data oriented,
8-9 people, many projects,
Growth of this role / level, visibility, opportunity,